



Sri Chaitanya IIT Academy.,India.

A.P. T.S. KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Sr. XL-500, Sr.S60(N&S) & LIIT

JEE-MAIN

Date: 21-12-2025

Time: 09.00Am to 12.00Pm

GTM-03

Max. Marks: 300

KEY SHEET

MATHEMATICS

1	3	2	2	3	4	4	1	5	2
6	3	7	3	8	2	9	3	10	4
11	4	12	3	13	4	14	1	15	2
16	3	17	1	18	1	19	1	20	2
21	5	22	100	23	0	24	1	25	2

PHYSICS

26	4	27	3	28	2	29	4	30	1
31	1	32	4	33	3	34	3	35	2
36	2	37	1	38	4	39	3	40	4
41	3	42	1	43	4	44	4	45	1
46	4	47	6	48	5	49	2	50	5

CHEMISTRY

51	3	52	1	53	2	54	4	55	3
56	3	57	2	58	1	59	2	60	2
61	1	62	1	63	3	64	2	65	3
66	4	67	1	68	2	69	4	70	3
71	1	72	10	73	9	74	2	75	4



SOLUTION

MATHEMATICS

1. Find $P(x=1)+P(x=2)$

$$P(X \leq 2) = P(x=1) + P(x=2)$$

$$= \frac{{}^{12}C_1 \times {}^{40}C_1}{{}^{52}C_2} + \frac{{}^{12}C_2}{{}^{52}C_2} = \frac{6 \times 91}{51 \times 26} = \frac{7}{17}$$

2.

$$3. \left(\lim_{n \rightarrow \infty} \left(\frac{n!}{n^n} \right)^{1/n} \right) \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{3n} \frac{r/n}{\sqrt{16+(r/n)^2}} = \left(\frac{1}{e} \right) \int_0^3 \frac{x}{\sqrt{16+x^2}} dx = \frac{1}{e}$$

4. $f(x)=0$ has minimum 2 solutions in $[0,4]$

$$g(x)=0 \text{ has minimum 3 solutions in } [0,4]$$

$$g'(x)=0 \text{ has minimum 2 solutions in } [0,4]$$

$$\frac{d}{dx}(f(x)g'(x))=0 \text{ has minimum 3 solutions in } [0,4]$$

5. $k=2, a>3$

6. Centre of hyperbola is (3,4) one end of transverse axis is (3,-1)

$$\therefore \text{Transverse axis is parallel to y-axis} \Rightarrow b=5 \text{ and } b^2 - a^2 = 20 \Rightarrow a = \sqrt{5}$$

$$\therefore \text{Equation of hyperbola is } \frac{(x-3)^2}{5} - \frac{(y-4)^2}{25} = -1$$

7. Add and subtract

$$\begin{aligned} & 7!, 7!+8!+\frac{9!}{2!}+\frac{10!}{3!}+\dots+\frac{27!}{20!}-7!=7!\left(1+\frac{8!}{1!7!}+\frac{9!}{2!7!}+\frac{10!}{3!7!}+\dots+\frac{27!}{20!7!}-7!\right) \\ & 7!\left({}^7C_0+8C_1+9C_2+10C_3+\dots+{}^{27}C_{20}\right)-7!\left(8C_0+8C_1+9C_2+10C_3+\dots+{}^{27}C_{20}\right)-7! \\ & =7!\left({}^{28}C_{20}\right)-7! \end{aligned}$$

8. $y=mx+\frac{4}{m}$ and $hx-ky=h^2-k^2$ represents same line

9. The circles touch each other at (1,1)

$$10. x=\frac{1}{2}, y=\frac{1}{4} \Rightarrow k=\frac{1}{4} \cdot \frac{\sqrt{3}}{2} - \frac{1}{2} \cdot \frac{\sqrt{15}}{4} = \frac{\sqrt{3}-\sqrt{15}}{8}$$

$$y\sqrt{1-x^2}=k+x\sqrt{1-y^2}$$

$$\Rightarrow y \cdot \frac{1 \cdot (-2x)}{2\sqrt{1-x^2}} + y' \cdot \sqrt{1-x^2} = 1 \cdot \sqrt{1-y^2} + \frac{x \cdot (-2y)}{2\sqrt{1-y^2}} y'$$



$$\Rightarrow -\frac{xy}{\sqrt{1-x^2}} + y'\sqrt{1-x^2} = \sqrt{1-y^2} - \frac{xy \cdot y'}{\sqrt{1-y^2}}$$

$$\Rightarrow y' \left(\sqrt{1-x^2} + \frac{xy}{\sqrt{1-y^2}} \right) = \frac{xy}{\sqrt{1-x^2}} + \sqrt{1-y^2}$$

$$y' \left(\frac{\sqrt{3}}{2} + \frac{1}{8 \cdot \frac{\sqrt{15}}{4}} \right) = \frac{1}{8 \cdot \frac{\sqrt{3}}{2}} + \frac{\sqrt{15}}{4}$$

$$y' \left(\frac{\sqrt{45}+1}{2\sqrt{15}} \right) = \frac{(1+\sqrt{45})}{4\sqrt{3}} \Rightarrow y' = \frac{\sqrt{5}}{2}$$

$$11. \quad A = \int_0^{\frac{1}{3}} \sqrt{12x} dx + \int_{\frac{1}{3}}^1 \left(\frac{1}{x} - 1 \right) dx$$

$$12. \quad 4! \cdot (2!3 + 2 \times 2 + 3) = 312$$

$$13. \quad \frac{dx}{dt} = e^t \left\{ \frac{2 \tan t + 1 - \tan^2 t}{\sec^2 t} \right\} \quad \frac{dx}{dt} = e^t \{ \sin 2t + \cos 2t \}$$

$$\frac{dy}{dt} = e^t \left\{ \frac{1 - \tan^2 t - 2 \tan t}{(1 + \tan^2 t)} \right\} = e^t \{ \cos 2t - \sin 2t \}$$

$$\frac{dy}{dx} = \frac{\cos 2t - \sin 2t}{\cos 2t + \sin 2t} \Rightarrow \frac{dy}{dx} \bigg|_{t=\frac{\pi}{4}} = -1$$

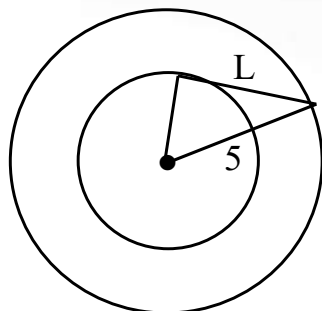
14. A) Domain of $f(x)$ is $[-2, 2]$.

$$f(x)|_{\min} = 0 + 4 - 8 + 1 = -3 \quad \text{at } x = 2$$

$$f(x)|_{\max} = \pi + 4 + 8 + 1 = \pi + 13 \text{ at } x = -2$$

$$\therefore \text{Range is } [-3, \pi + 13] \quad a + b = -3 + 13 = 10$$

B)



$$L = \sqrt{25 - 21} = 2$$



$$C) \int_0^6 e^{\{3x\}} dx = \int_0^{6.3\frac{1}{3}} e^{\{3x\}} dx = 18 \int_0^{1/3} e^{3x} dx = 18 \left(\frac{e^{3x}}{3} \right)_0^{1/3} = 6(e-1)$$

$$\therefore p+q=6+1=7$$

$$D) x^2 - (1+b)x + b - 2 = 0$$

$$f(1)=1-1-b+b-2=-2 \quad \therefore f(x)=0 \text{ has atleast one positive root.}$$

$$15. \quad T_6 = ar^5 = 96 \quad a = \frac{96}{r^5}$$

$$500 < T_n < 780 \Rightarrow 500 < ar^{(n-1)} < 780$$

$$\Rightarrow 500 < 96r^{n-6} < 780 \Rightarrow \frac{500}{96} < r^{(n-6)} < \frac{780}{96} \Rightarrow 5.2 < r^{(n-6)} < 8.1$$

$$r^{n-6} = 6; r^{n-6} = 7; r^{n-6} = 8$$

$$r=6, n=7; r=7, n=7; r=8, n=7, r=2, n=9$$

For $r = 6, 7, 8$

$$0 < a < 1 \Rightarrow [a] = 0$$

$$r=2 \text{ is possible } \quad n-6=3 \quad n=9$$

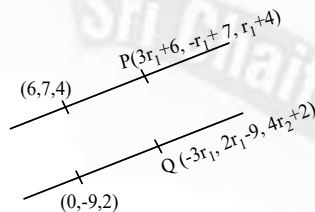
$$16. \quad V_1(\vec{a} + \vec{b}) \cdot ((\vec{b} + \vec{c}) \times (\vec{c} + \vec{a})) = 2[\vec{a} \vec{b} \vec{c}]$$

$$V_2 = \frac{1}{6}(\vec{a} \times \vec{b}) \cdot [(\vec{b} \times \vec{c}) \times (\vec{c} \times \vec{a})]$$

$$= \frac{1}{6}(\vec{a} \times \vec{b}) \cdot [((\vec{b} \times \vec{c}) \cdot \vec{a})\vec{c} - ((\vec{b} \times \vec{c}) \cdot \vec{c})\vec{a}] = \frac{1}{6}[\vec{a} \vec{b} \vec{c}]^2$$

$$[\vec{a} \vec{b} \vec{c}]^2 = \begin{vmatrix} -3 & 1 & 1 \\ 4 & 2 & 4 \\ 2 & 2 & 0 \end{vmatrix} = 36 \quad \therefore V_1 = 76 \text{ and } V_2 = 216$$

17.



direction ratio of

$$PQ = (-3r_2 - 3r_1 - 6, 2r_2 + r_1 - 16, 4r_2 - r_1 - 2)$$

$$\text{Now, } -3(-3r_2 - 3r_1 - 6) + 2(2r_2 + r_1 - 16) + 4(4r_2 - r_1 - 2) = 0$$

$$7r_1 + 29r_2 = 22$$

$$3(-3r_2 - 3r_1 - 6) - (2r_2 + r_1 - 16) + (4r_2 - r_1 - 2) = 0$$

$$7r_1 + 29r_2 = 22 \quad \dots\dots\dots(1)$$



$$11r_1 + 7r_2 = -4 \quad \dots\dots(2)$$

On solving (1) & (2)

$$r_1 = -1, r_2 = 1 \quad \text{So } P(3, 8, 3)$$

Image of $P(3, 8, 3)$ w.r.t plane $3x + 3y - z = 11$. $R(-3, 2, 5) \quad a + b + c = 4$

$$\bullet P(3, 8, 3)$$



$$\bullet P(-3, 2, 5)$$

$$18. \quad 3^{\sin 2x + 2 \cos^2 x} + 3^{1 - \sin 2x + 2 \sin^2 x} = 28$$

$$1 - \sin 2x + 2 \sin^2 x = 1 - (\sin 2x - 2 \sin^2 x)$$

$$= 3 - (\sin 2x + 2 \cos^2 x)$$

$$\therefore 3^{\sin 2x + 2 \cos^2 x} + 3^{3 - (\sin 2x + 2 \cos^2 x)} = 28$$

$$\text{Put } 3^{\sin 2x + 2 \cos^2 x} = t \quad t + \frac{27}{t} = 28$$

$$t^2 - 28t + 27 = 0 \quad t = 1, 27 \quad \therefore \sin 2x + 2 \cos^2 x = 0, 3$$

$$\Rightarrow 2 \sin x \cos x + 2 \cos^2 x = 0, 3$$

$$\text{Case-I: } 2 \cos x (\sin x + \cos x) = 0 \quad \Rightarrow \cos x = 0, \tan x = -1$$

$$\text{Case-II: } 2 \cos x (\sin x + \cos x) = 3 \quad \Rightarrow \cos 2x + \sin 2x = 2 \text{ Which is not}$$

$$\text{From case-I general solution is } x = (2K + 1)\frac{\pi}{2}, K\pi - \frac{\pi}{4}, K \in I$$

$$19. \quad -\pi \leq \pi \sin x \leq \pi \quad \Rightarrow -1 \leq \cos(\pi \sin x) \leq 1$$

$$\Rightarrow -1 \leq \sin\left(\frac{\pi}{2}(\cos \pi \sin x)\right) \leq 1 \quad \Rightarrow -\pi \leq \pi \sin\left(\frac{\pi}{2} \cos(\pi \sin x)\right) \leq \pi \quad \Rightarrow -1 \leq y \leq 1$$

$$20. \quad a R b \Leftrightarrow a = 2^k \cdot a \text{ it is true for } k = 0 \quad \therefore \text{ reflexive}$$

$$(2, 1) \in R \text{ but } (1, 2) \notin R \Rightarrow \text{it is not symmetric}$$

$$\text{If } a = 2^{k_1} b \text{ and } b = 2^{k_2} c, \text{ then } a = 2^{k_1 + k_2} c \Rightarrow \text{it is transitive}$$

$$21. \quad \text{Given the system of equation}$$

$$x + y - z = 2 \quad \dots(1)$$

$$2x - y + 4z = 1 \quad \dots\dots(2)$$

$$x + \alpha y + z = \beta \quad \dots\dots(3)$$

$$\text{Solving (1) and (2)} \quad : x = 1 - z, y = 1 + 2z$$

$$\therefore (3) \Rightarrow 1 - z + \alpha(1 + 2z) + z = \beta \quad \Rightarrow 2\alpha z = \beta - \alpha - 1$$

$$\therefore \text{For unique solution } \alpha \neq 0, \beta \in R \quad \therefore A = 4 \times 5 = 20$$

$$\text{For no solution } \alpha = 0, \beta \neq 1 \quad \Rightarrow B = 1 \times 4 = 4$$

and for infinite number of solutions,

$$\alpha = 0, \beta = 1 \Rightarrow C = 1 \quad \therefore A - 4B + C = 20 - 16 + 1 = 5$$



$$22. A = \begin{bmatrix} -1 & a \\ 0 & b \end{bmatrix} \quad A^2 = \begin{bmatrix} -1 & a \\ 0 & b \end{bmatrix} \begin{bmatrix} -1 & a \\ 0 & b \end{bmatrix} = \begin{bmatrix} 1 & -a+ab \\ 0 & b^2 \end{bmatrix}$$

$$\therefore T_n = \left\{ A \in S; A^{n(n+1)} = I \right\}$$

$\therefore b$ must be equal to 1

$\therefore$ In this case A^2 will become identity matrix and

A can take any value from 1 to 100

$\therefore$ Total number of common element will be 100

$$23. \lim_{x \rightarrow 2p^+} \left(\frac{1 - \cos(x^2 - 4px + q^2 + 8q + 16)}{(x^2 - 4px + q^2 + 8q + 16)^2} \right) \left(\frac{(x^2 - 4px + q^2 + 8q + 16)^2}{(x - 2p)^2} \right)$$

$$\lim_{h \rightarrow 0} \frac{1}{2} \left(\frac{(2p+h)^2 - 4p(2p+h) + q^2 + 8q + 16}{h^2} \right) = \frac{1}{2}$$

Using L'Hospital's

$$\lim_{x \rightarrow 2p^+} [f(x)] = 0 \quad 24. \text{Sol: } \sigma^2 = \frac{\sum x_i^2}{N} \left(\frac{\sum x_i}{N} \right)^2$$

$$\sum (2x_i + 1)^2 = 21n \quad \Rightarrow 4A + n + 4B = 29n \quad \Rightarrow A + B = 7n$$

$$\text{Where } A = \sum_{i=1}^n x_i^2, B = \sum_{i=1}^n x_i \quad \sum (3x_i - 1)^2 = 34n \quad \Rightarrow 9A + n - 6B = 34n$$

$$\Rightarrow 3A - 2B = 11n \quad \therefore A = 5n, B = 2n \quad \sigma^2 = \frac{1}{n} \frac{\sum x_i^2}{N} \left(\frac{1}{n} \frac{\sum x_i}{N} \right)^2 = \frac{A}{n} - \frac{B^2}{n^2} = 5 - 4 = 1$$

$$25. f(x) = |(x-3)(x+1)| \cdot e^{(3x-2)^2}$$

$$f(x) = \begin{cases} (x-3)(x+1) \cdot e^{(3x-2)^2} & ; \quad x \in (3, \infty) \\ -(x-3)(x+1) \cdot e^{(3x-2)^2} & ; \quad x \in [-1, 3] \\ (x-3) \cdot (x+1) \cdot e^{(3x-2)^2} & ; \quad x \in (-\infty, -1) \end{cases}$$

Clearly non-differentiable at $x = -1$ & $x = 3$.

PHYSICS



26. $y = \beta x^2 \Rightarrow V_y = 2\beta x V_x = 2\beta V_x (V_x t) \text{ (as } x = V_x t)$

Similarly $V_y = \alpha t \Rightarrow \alpha t = 2\beta V_x^2 t \text{ and } \sqrt{\frac{\alpha}{2\beta}}$

27. If A moves down the incline by 1 metre, B shall move up by $\frac{1}{2}$ metre. If the speed of B is v then the speed of A will be $2v$.

From conservation of energy:

Gain in K.E. = loss in P.E.

$$\frac{1}{2} m_A (2v)^2 + \frac{1}{2} m_B v^2 = m_A g \times \frac{3}{5} - m_B g \times \frac{1}{2}$$

Solving we get $v = \frac{1}{2} \sqrt{\frac{g}{3}}$

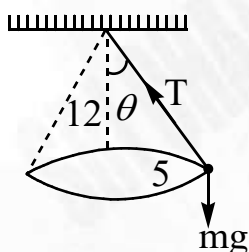
28. $mg \sin \theta_2 = mg \sin \theta_1 + \mu mg \cos \theta$

29. $dw = \vec{F} \cdot d\vec{r} = \vec{F} \cdot (dx\hat{i} + dy\hat{j})$

$$= K \int \frac{xdx}{(x^2 + y^2)^{3/2}} + \frac{ydy}{(x^2 + y^2)^{3/2}}$$

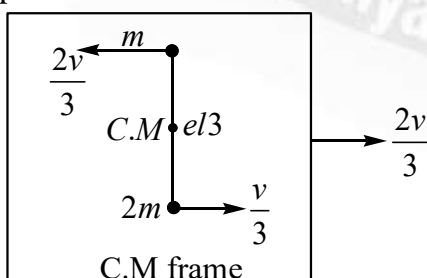
$$x^2 + y^2 = a^2 \quad w = \frac{k}{a^3} \int_a^0 x dx + \int_0^a y dy = \frac{k}{a^3} \left(\frac{-a^2}{2} + \frac{a^2}{2} \right) = 0$$

30.



$$T \cos \theta = mg \quad T \sin \theta = mr\omega^2 \quad \text{Time taken} = \frac{2\pi}{\omega}$$

31. Solve the problem in CM frame. In CM frame particles A and B moving along circular path with same ω



Velocity of 2m mass $= v - \frac{2v}{3} = \frac{v}{3}$

Angular velocity of particle A



$$\omega = \frac{v/3}{l/3} = \frac{v}{l} \quad T = M\omega^2 r = 2m \cdot \frac{v^2}{l^2} \cdot \frac{l}{3} \quad \text{Or } T = \frac{2mv^2}{3l}$$

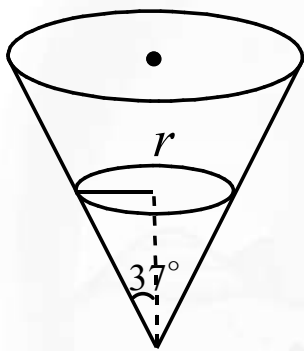
32. Let v and ω be the linear speeds of the rod after applying an impulse J and B .
The form: Impulse = change in momentum

We have $mv = J$ or $v = \frac{J}{m}$ $I\omega = J \cdot \frac{1}{2}$ Or $\frac{ml^2}{12} \cdot \omega = J \cdot \frac{1}{2}$ or $\omega = \frac{6J}{ml}$

After the given time $t = \frac{\pi ml}{12J}$, the rod will rotate an angle $\theta = \omega t = \left(\frac{6J}{ml}\right) \left(\frac{\pi ml}{12J}\right) = \frac{\pi}{2}$

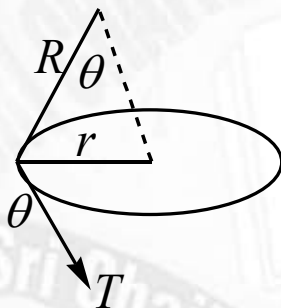
$$\frac{1}{2} \cdot \omega = \left(\frac{l}{6}\right) \left(\frac{6J}{ml}\right) = \frac{J}{m} = v \therefore |\vec{v}_p| = \sqrt{2}v = \sqrt{2} \frac{J}{m}$$

33. $\frac{r}{8} = \tan 37^\circ = \frac{3}{4}$



$$F = (P_0 + h\rho g)\pi r^2 = 2 \times 10^7 N$$

34.



Surface tension force $= (T \sin \theta) \times 2\pi Tr$ ($\because r = R \sin \theta$)

$$= T \sin \theta \times 2\pi R \sin \theta = 2\pi RT \sin^2 \theta$$

35. Temperature of sun's core is $T = \frac{b}{\lambda_m}$

$$\text{Rate of energy radiated by sun} = \sigma (4\pi R^2) \left(\frac{b}{\lambda_m}\right)^4 = \frac{\Delta m}{t} c^2$$

$$= \frac{\text{net energy obtained from conversion of mass of Hydrgen in sun's core}}{\text{total time}}$$



$$= \frac{(\eta m)c^2}{t} \Rightarrow t = \frac{\eta mc^2 \lambda_m^4}{\sigma(4\pi R^2)b^4}$$

36. $f_0 = \frac{1}{2\pi} \sqrt{\frac{mgl}{I}}$

Where, l is distance between point of suspension and centre of mass of the body thus, for the stick of length L and mass m :

$$f_0 = \frac{1}{2\pi} \sqrt{\frac{mg \cdot \frac{L}{2}}{\left(\frac{mL^2}{3}\right)}} = \frac{1}{2\pi} \sqrt{\frac{3g}{2L}}$$

When bottom half of the stick is cut off $f_0' = \frac{1}{2\pi} \sqrt{\frac{\frac{m}{2} \cdot g \cdot \frac{L}{4}}{\frac{m(L/2)^2}{3}}} = \frac{1}{2\pi} \times \sqrt{\frac{3g}{L}} = \sqrt{2} f_0$.

37. Force experienced by dipole in non-uniform electric field, is given by $F = P \frac{\partial E}{\partial l}$.

Here, $F = \frac{Q}{4\pi\epsilon_0 r^2}$

We have to take derivative of E along dipole length, i.e., along $\vec{r}$.

So, $\frac{\partial E}{\partial r} = -\frac{2Q}{4\pi\epsilon_0 r^3}$

38. $\frac{1}{C_{eq}} = \frac{1}{8} + \frac{1}{24} + \frac{1}{12} \Rightarrow C_{eq} = 4\mu F \quad q = C_{eq}(V_A - V_B) = 240\mu C$

39. Path is helical. In a reference frame travelling at $v \cos \theta$, path will be seen circular. This is an inertial frame so acceleration should be same

$$\Rightarrow \frac{(v \sin \theta)^2}{R'} = \frac{v^2}{r} \Rightarrow r = R' \cos \theta$$

but $R' = \frac{mv \sin \theta}{Bq} = R \sin \theta \Rightarrow r = R \cos \theta$

40. Conceptual

41. $\phi_A = \frac{\mu_0 i \pi R^2}{2\pi(R^2 + x^2)^{3/2}} \cdot \pi r^2$

$$B_A = \frac{d\phi}{dt} = \frac{\mu_0 i \pi}{2} R^2 r^2 \left(\frac{-3}{2}\right) (R^2 + x^2)^{-5/2} \cdot 2x$$

B_A is maximum when $\frac{dB_A}{dx} = 0 \Rightarrow \frac{d}{dx} \frac{x}{(R^2 + x^2)^{5/2}} = 0$



$$\text{Or } (R^2 + x^2)^{5\beta} = \frac{5x}{2}(R^2 + x^2)^{3\beta} \quad 2x = 0 \quad \text{Or } R^2 + x^2 - 5x^2 = 0 \quad \text{or } x = \frac{R}{2}$$

42. Voltage in primary coil $V_p = 200V$

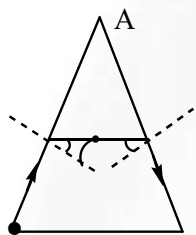
Current in primary coil $I_p = 2A$

Voltage in secondary coil $V_s = 2000V$

Current in secondary coil $I_s = ?$

$$\frac{V_s}{V_p} = \frac{I_p}{I_s} \quad \frac{2000}{200} = \frac{2}{I_s} \quad I_s = 0.2A$$

43. To satisfy the conditions



$$r = C \text{ and } r' = C \quad \therefore A = r + r' = 2C = 2 \sin^{-1}(2/3)$$

44. The binding energy per nucleon is independent of atomic number for medium to large nuclei. As, the distance between nucleon increase, nuclear force decreases in case of medium to large nuclei. This is due to short range of nuclear force. Hence, both A and R are true and R is the correct explanation of A.

45. Conceptual

46. The given wave is superposition of 3 waves with frequency, ω_0 , $\omega + \omega_0$ and $\omega_0 - \omega$.

$$\omega_{\max} = (\omega_0 + \omega). \quad \therefore E_{\max} = h\nu_{\max} = h \frac{(\omega_0 + \omega)}{2\pi}$$

$$h\nu_{\max} = KE_{\max} + \phi \Rightarrow \phi = 4eV$$

47.
$$\Delta E = 13.6Z^2 \left[\frac{1}{n_1} - \frac{1}{n_2} \right]$$

For the transition from a higher state n to the first excited state $n_1 = 2$ and $n_2 = n$ the total energy released is $(10.2 + 17.0)eV = 27.2eV$. Thus, $\Delta E = 27.2eV$, $n_1 = 2$.

$$(i) \therefore 27.2 = 13.6Z^2 \left[\frac{1}{4} - \frac{1}{n^2} \right]$$

For the eventual transition to the second excited state $n_1 = 3$ and $n_2 = n$ the total energy released is $(4.25 + 5.95)eV$ or $10.2eV$. Thus, (ii) $10.2 = 13.6Z^2 \left[\frac{1}{9} - \frac{1}{n^2} \right]$

Solving eqns (i) and (ii), we get: $n = 6$ and $Z = 3$

48. Least count of screw gauge $= \frac{1}{100} mm = 0.01mm$

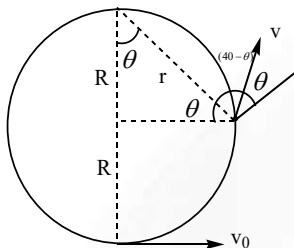
Diameter of the wire $= (1 + 25 \times 0.01)mm = 0.125$



$$\text{Since } Y = \frac{4Tl}{\pi d^2 \delta l} \quad \therefore \frac{\Delta Y}{Y} = \frac{\Delta l}{l} + \frac{2\Delta d}{d} + \frac{\Delta(\delta l)}{\delta l} = \frac{0.01}{50} + \frac{2 \times 0.001}{0.125} + \frac{0.001}{0.125} = 0.0242.$$

$$\text{Percentage error} = \frac{\Delta Y}{Y} \times 100 = 2.42 \quad \therefore n = 5$$

49.



$$(v \cos \theta) 2R \cos \theta = 2Rv_0 \quad \dots\dots(1)$$

$$v = \frac{v_0}{\cos^2 \theta}$$

$$\frac{mv^2}{R} = F \cos \theta \quad \dots\dots(2)$$

$$F = \left(\frac{m}{R \cos \theta} \right) \frac{v_0^2}{\cos^4 \theta} \quad F = \frac{mv_0^2}{R \cos^5 \theta}$$

$$r = 2R \cos \theta \quad \dots\dots(3)$$

$$\Rightarrow F = \frac{mv_0^2}{R \left(\frac{r}{2R} \right)^5} = \frac{(2^5)mv_0^2 R^4}{r^5} \quad F = \frac{32mv_0^2 R^4}{r^5}$$

$$x=1, y=4, z=2, \text{ and } k=5$$

$$x+y+z-k=1+4+2-5=2$$

$$50. \quad \eta = \frac{\text{Total work done}}{\text{Total heat supplied}}$$

$$W_{AB} = 5000J$$

$$W_{BC} = 1200 R \ln 2$$

$$Q_{AB} = nC_p \Delta T = 2 \times \frac{5}{2} R \times 300 = 1500R$$

$$Q_{BC} = W_{BC} = 1200R \ln 2$$

$$\eta = \frac{600R(2 \ln 2 - 1)}{1800R + 1200R \ln 2} = 1 - \frac{21}{15 + 12 \ln 2} \quad x = 7$$

CHEMISTRY

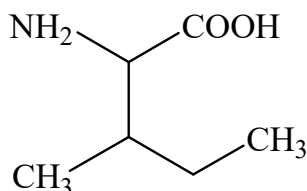
51. Carboxylic acid is more reactive than ketone for Borane reduction

52. Conceptual

53. Intramolecular aldol reaction followed by iodoform reaction

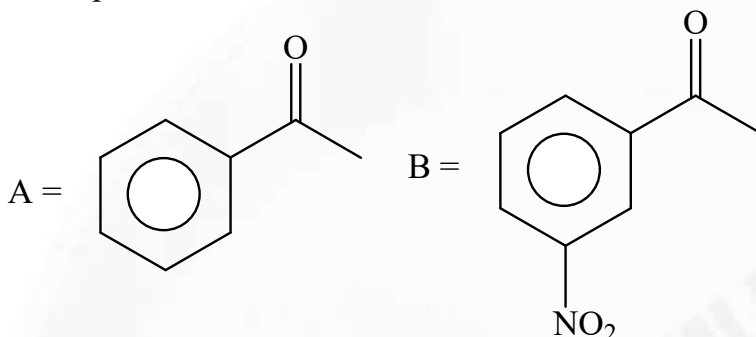


54.



55. Endorphin hormone is made up of poly peptide.

56. Conceptual



57.

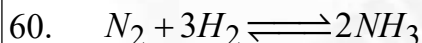
In clemenson reduction nitro group is disturbed

58. let V ml AlCl_3 is needed

$$0.6 = \frac{2.4V + 20}{V + 50}$$

$$V = 5.56 \text{ ml}$$

59. Conceptual



1-x 3-3x 2x at equilibrium

gm.eq of NH_3 = gm eq of H_2SO_4

$$2x = 1 \times 2 \times 0.5$$

$$x = 1/2$$

$$[\text{NH}_3] = 0.5 \text{ mole/l}$$

$$[\text{H}_2] = 1.5 \text{ mole/l}$$

$$[\text{NH}_3] = 1 \text{ mole/l}$$

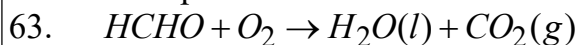
$$K_c = \frac{(1)^2}{0.5 \times (1.5)^3} = 0.59$$

61. At freezing point

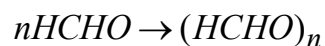
$$12 - \frac{3000}{T} = 6 - \frac{2400}{T}$$

$$6 = \frac{600}{T} \quad T = 100 \text{ Kelvin}$$

62. conceptual

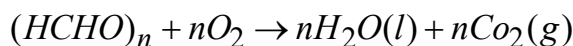


$$\Delta H = -134$$





$$\Delta H = -72$$



$$\Delta H = -732$$

$$-134n + 72 = -732$$

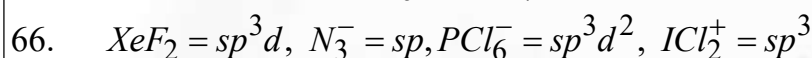
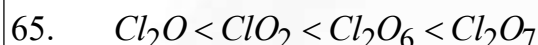
$$n = 6$$

Formula of paraformaldehyde is $C_6H_{12}O_6$

$$64. \Delta G_{R \times n}^0 = [2 \times (-237.2) + 1 \times (-394.4)] - [-166.2] \\ = -702.6 \text{ kJ / mole}$$

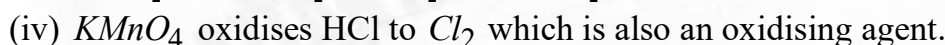
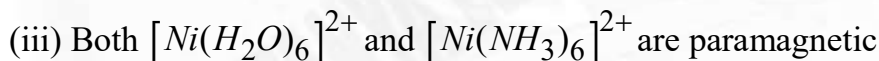
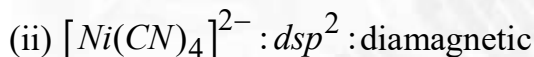
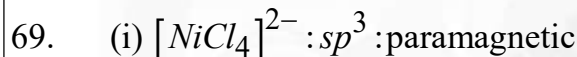
$$\Delta H_{comb}^0 = -826 \text{ kJ / mole}$$

$$\% \eta = \frac{\Delta G^0}{\Delta H^0} \times 100 = \frac{-702.6}{-826} \times 100 = 85\%$$



68. Highest fluoride of Mn : MnF_4 .

Highest fluoride of Cr : CrF_6 .



70. (i) Most of the reactions of fluorine are exothermic due to the small and strong bond formed by it with other elements

(ii) oxoacid of fluorine is HOF

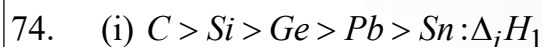
71. Dye-azo test

$$72. x = 6, y = 1, w = 2, q = 1$$

$$73. \frac{l}{A} = \frac{2}{4}; R = \rho \cdot \frac{1}{2} \text{ or } \rho = 200\Omega \quad \kappa = \frac{1}{\rho} = \frac{1}{200} \Omega^{-1} cm^{-1}$$

$$\Lambda = \frac{1}{200} \times \frac{1000}{0.1} = 50 \quad \Lambda^0 = 300 + 100 = 400 \quad \alpha = \frac{\Lambda}{\Lambda^0} = \frac{50}{400} = \frac{1}{8}$$

$$\Delta T_b = i \times K_b \times m \quad \Delta T_b = \frac{9}{8} \times 0.8 \times 0.1 = 9 \times 10^{-2}$$



(ii) stable oxidation state of lead is +2

